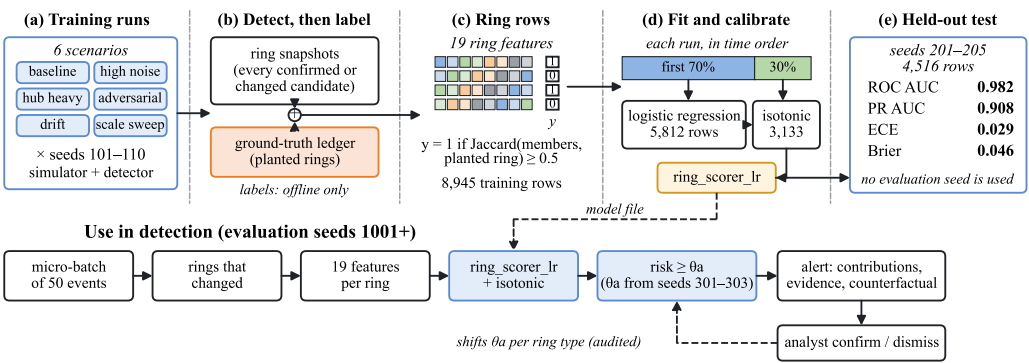

Training (offline, labelled seeds only)
(a) Training runs
6 scenarios
baseline
high noise
hub heavy
adversarial
drift
scale sweep
× seeds 101–110
simulator + detector
(b) Detect, then label
ring snapshots
(every confirmed or
changed candidate)
ground-truth ledger
(planted rings)
labels: offline only
(c) Ring rows
19 ring features
y
y = 1 if Jaccard(members,
planted ring) ≥ 0.5
8,945 training rows
(d) Fit and calibrate
each run, in time order
first 70%
30%
logistic regression
5,812 rows
isotonic
3,133
ring_scorer_lr
(e) Held-out test
seeds 201–205
4,516 rows
ROC AUC 0.982
PR AUC 0.908
ECE 0.029
Brier 0.046
no evaluation seed is used
model file
Use in detection (evaluation seeds 1001+)
micro-batch
of 50 events
rings that
changed
19 features
per ring
ring_scorer_lr
+ isotonic
risk ≥ θa
(θa from seeds 301–303)
alert: contributions,
evidence, counterfactual
analyst confirm / dismiss
shifts θa per ring type (audited)